Physics of the Earth and Planetary Interiors, 43 (1986) 148–159
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

Palaeomagnetism of porphyritic stocks and rhyolite sills, Avalon Zone of eastern Newfoundland

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(Received July 4, 1985; revision accepted January 21, 1986)

Seguin, M.K.- and Lortie, J., 1986. Palaeomagnetism of porphyritic stocks and rhyolite sills, Avalon Zone of eastern Newfoundland. Phys. Earth Planet. Inter., 43: 148–159.

The eastern segment of the Appalachian orogen is largely underlain by late Precambrian (Hadrynian) rocks affected by the Avalonian, Acadian and possibly Alleghenian orogenies. The provenance of the Avalon Zone of Newfoundland is uncertain. The region investigated in this segment consists of porphyrite stocks and sills (laccoliths) intrusive into the sedimentary, tuffaceous and volcanic rocks of the Harbour Main Group and rhyolite sills intrusive into the porphyrites. Some 55 oriented samples (148 specimens) collected at 11 sites were thermally (20–650°C) and AF (0.05–100 mT) demagnetized. Three components of magnetization were isolated: C (311°, +48°, $\alpha_{95} = 11°$, $k = 21$, 10 sites), A (13°, +37°, $\alpha_{95} = 14°$, $k = 22$, 6 sites), and B (67°, +45°, $\alpha_{95} = 15°$, $k = 27{,}5$ sites). Based on coercivity spectra, unblocking temperatures, frequency distribution and precision parameters of the respective components, it is suggested that component C is older than component A which is turn is older than component B. The palaeopoles of components C, A and B are: 211°E, 48°N ($d_p = 9.8°$, $d_m = 14.7°$); 101°E, 61°N ($d_p = 9.6°$, $d_m = 16.4°$); 33°E, 34°N ($d_p = 12°$, $d_m = 19°$), respectively. Component C is most probably primary. Component A is secondary and its pole is near that of Carboniferous and Early Permian North America poles, indicating that the porphyrites and the rhyolites were remagnetized in the late Palaeozoic. Component B remains unexplained; it is possible that it is an unresolved pseudo-component but it is more likely an overprint. There are few palaeomagnetic results for the late Precambrian period in Avalon terrane(s). The preliminary results of this study suggest the presence of a separate plate from North America at that time. These results will prove useful for the palaeoreconstruction of the continents (North Africa, northeast Europe) in the late Precambrian period.

1. Introduction

The eastern segment of the Appalachian orogen is largely underlain by late Precambrian (Hadrynian) rocks. Systematic mapping programs of the Avalon Zone were first undertaken by Rose (1952) and are still continuing (e.g., Nixon and Papezik, 1979). The Precambrian stratigraphy was subsequently elucidated by Rodgers (1967), Williams and King (1979) and others.

The Avalon Peninsula was affected by two principal orogenies, a late Precambrian 'Avalon orogeny' (Lilly, 1966) and the 'Acadian' orogeny of Middle to Late Devonian age (e.g., Dallmeyer et al., 1983). Different tectonic models have been proposed for the origin of the Avalon mini-continent: (1) island arc (e.g., Hughes and Brückner, 1971), and ensialic island arc (Rast et al., 1976; Rast and Skehan, 1983); (2) subduction and basin and range-type rifting (e.g., Papezik, 1970; Schenk, 1978); and (3) continental extension and rifting (Rankin, 1976). In recent models, the Avalon Zone is regarded as a suspect terrane and exotic to North America (Williams and Hatcher, 1982, 1983).

Precambrian rocks similar to those encountered

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Fig. 1. General geology of the northern Avalon sector of Newfoundland and location of the study area.

in the Avalon Zone occur throughout the Appalachian orogen in Nova Scotia, New Brunswick, eastern Massachusetts, southeastern United States (Carolina Slate, Raleigh, Eastern Slate, Kiokee belts (e.g., Schenk, 1971; Williams, 1979)), and on the eastern side of the Atlantic, i.e., Wales, Brittany, the Iberian Peninsula, Czechoslovakia and Turkey as well as in northeast Africa (e.g., Choubert, 1935; Cogné, 1972; Strong, 1979).

The provenance of the Avalon Zone of Newfoundland is still uncertain. King (1979) observed a close resemblance of the Precambrian sedimentary units of the Avalon Peninsula to the assemblages in the Brioverian of France and the lower Dalradian of Scotland. In this model, the Avalon Zone was an original part of the North European–Baltic plate (see also Deutsch, 1984). Hagstrum et al. (1980) and Perroud et al. (1983) suggested that the Avalonian terranes of northeast North America were part of a large lithospheric plate called 'Armorica' throughout much of Palaeozoic time along with England and Wales as well as Brittany. O'Brien et al. (1983) suggested that the Precambrian sequences of the Avalon Zone in Canada are a segment of a Pan-African orogenic belt incorporated into the Appalachian Orogen during Palaeozoic times as its southeastern margin. The Avalon Zone would then be related to the Pan-African belt adjacent to the West African craton. The main objective of this

palaeomagnetic study and others in progress is to construct a reliable APWP for the late Precambrian period in Avalonian terrane(s).

2. Geological setting

The general geological setting of late Precambrian rocks of the Avalon Peninsula, Newfoundland, was described by King (1979) and O'Brien et al. (1983). Detailed lithostratigraphic, petrographic, petrochemical and tectonic studies on late Precambrian (Hadrynian) sediments and flow tuffs, volcanic rocks, ignimbrites and associated rocks of the Harbour Main Group in eastern Newfoundland were carried out by Papezik (1969, 1970, 1972), Hughes and Malpas (1971), Nixon (1974), and Nixon and Papezik (1979). The volcano-sedimentary sequence of the Harbour Main Group is associated with intrusive rocks. In the Colliers Bay area (Figs. 1 and 2), the most common intrusive rock unit is porphyrite (Nixon and Papezik, 1979) characterized by a large proportion (20–40%) of albite and smaller amounts of pyroxene, set in a matrix of divitrified and altered glass and iron oxide. The intrusive bodies are locally altered; the primary feldspar, which was compositionally zoned, was selectively replaced by epidote and the original pyroxene is often altered to chlorite. The glassy and microcrystalline matrix is often altered to chlorite, epidote, sericite, calcite, clay minerals and leucoxene (Hughes and Malpas, 1971; Nixon and Papezik, 1979).

The porphyrites contain variable but significant amounts of xenocrysts and xenoliths derived from the ash-flow tuffs, tuff-breccia, ignimbrites and volcanogenic sediments; the inclusions are most common along margins of the intrusive bodies. One striking feature of the porphyrites is the widespread occurrence of autoclastic intrusive breccia (Fisher, 1961) consisting of angular, almost equidimensional blocks of porphyrite surrounded by a fine groundmass with numerous xenocrysts; this probably resulted from a late magmatic pulse (Nixon, 1974; Nixon and Papezik, 1979).

The porphyrites are observed as irregular stocks and thick lens-like sills and dykes that may have

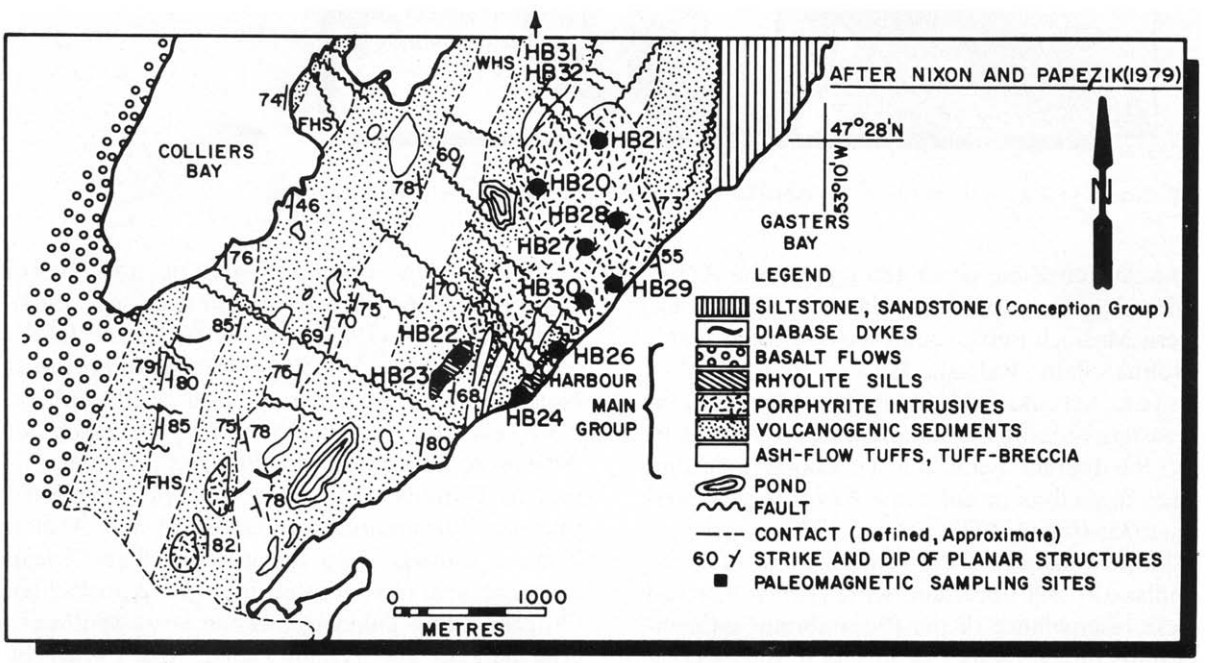


Fig. 2. Detailed geology of the study area (after Nixon and Papezik, 1979) and distribution of the palaeomagnetic sampling sites (HB20 to HB32) in porphyrite intrusives and rhyolite sills.

originally been laccoliths. These clearly cut the ash-flow tuffs, ignimbrites and intercalated volcanogenic sediments. However, small porphyrite clasts are found in the lithic fraction of the ash-flow tuffs and ignimbrites surrounding the Finn Hill and Weaver's Hill sequences, indicating that the intrusion of the porphyrites was roughly contemporaneous with the formation of the intrusive and sedimentary rocks (Nixon and Papezik, 1979). Two sills of light grey aphanitic rhyolite, some 400 m apart which intrude the southern part of the Bacon Cove ash-flow sequence (Fig. 2) were also sampled. The sills are 450 m long and 55–85 m thick. The rhyolites show conspicuous flow banding and consist of microphenocrysts of albite and quartz set in a microcrystalline quartzo-feldspathic groundmass with abundant fine-grained iron oxides. Fine-grained greenish-grey to black diabase dykes cut all other rocks on the Colliers Peninsula, including porphyrite and rhyolite sills (Nixon, 1974; Nixon and Papezik, 1979; Seguin et al., 1982). Most diabase dykes strike easterly, across the regional trend, and have steep dips.

3. Age

Anderson (1972) suggested an age range of 800–600 Ma for the Conception Group, considered to be derived from the Harbour Main Group (McCartney, 1967). Rast et al. (1976) and Strong (1979) correlated the Love Cove Group from the Burin Peninsula in southeastern Newfoundland with the Harbour Main Group. Stukas (1977) suggested an age of 660 ± 20 Ma ($^{40}Ar/^{39}Ar$ method) for the Belle Bay volcanics of the Avalon Peninsula. Rast et al. (1976) correlated the largest part of the Coldbrook Group volcanics with the Love Cove and Harbour Main Groups and assigned an age ranging from 750 to 640 Ma to the Coldbrook Group. Based on observed field relationships, McCartney (1967) considers the Holyrood Pluton to be a single intrusion post-dating the Harbour Main Group volcanics; the granitic portion of the pluton has been dated at 575 ± 11 Ma using the Rb–Sr method (609 ± 11 Ma using $^{87}Rb = 1.39 \times 10^{-11}$ years). Krogh et al. (1983) obtained an age of 621 ± 2 Ma (U/Pb). Considering the timing of this geological event, King (1979) suggested that the Harbour Main Group may be related in time to the Vendian of northern Eurasia. Since the geological relationships show that the intrusion of the porphyrites was slightly younger or roughly contemporaneous with the formation of the extrusive and sedimentary rocks of the Finn Hill, Weaver's Hill and Bacon Cove sequences, the porphyrite stocks and rhyolite sills of the Colliers Peninsula area have a possible age of 700–750 Ma. From a palaeomagnetic study of the Avalonian Finn Hill sequence of eastern Newfoundland, Seguin (1985) suggested an older age of about 730 Ma for this sequence.

4. Sampling procedure

Some 55 oriented hand samples (148 specimens) were collected at 11 different sites (seven in the porphyrite intrusions and four in the rhyolite sills); the orientation was taken with a Brunton or a sun compass. The number of samples per site was five and the number of specimens per sample between two and five. In the rhyolite sills, the majority of the samples were collected on the chilled margins of the sills. The oriented hand samples were later cored in the laboratory. The sample localities were chosen by reference to detailed geological maps (Nixon, 1974; Nixon and Papezik, 1979) where the structural geology was well understood. The locations of the sampling sites are shown on Fig. 2. No definite contact between the porphyrite intrusions and the Precambrian country rocks was observed on the field; consequently a baked contact test could not be performed.

5. Experimental procedure

The direction and intensity of permanent magnetization (NRM) were measured with a Schonstedt digital spinner magnetometer (model DSM-1) (sensitivity 10^{-4} Am^{-1}) coupled to PDP-11 and APPLE II PLUS minicomputers in tandem. The volume susceptibility (K) was measured at room

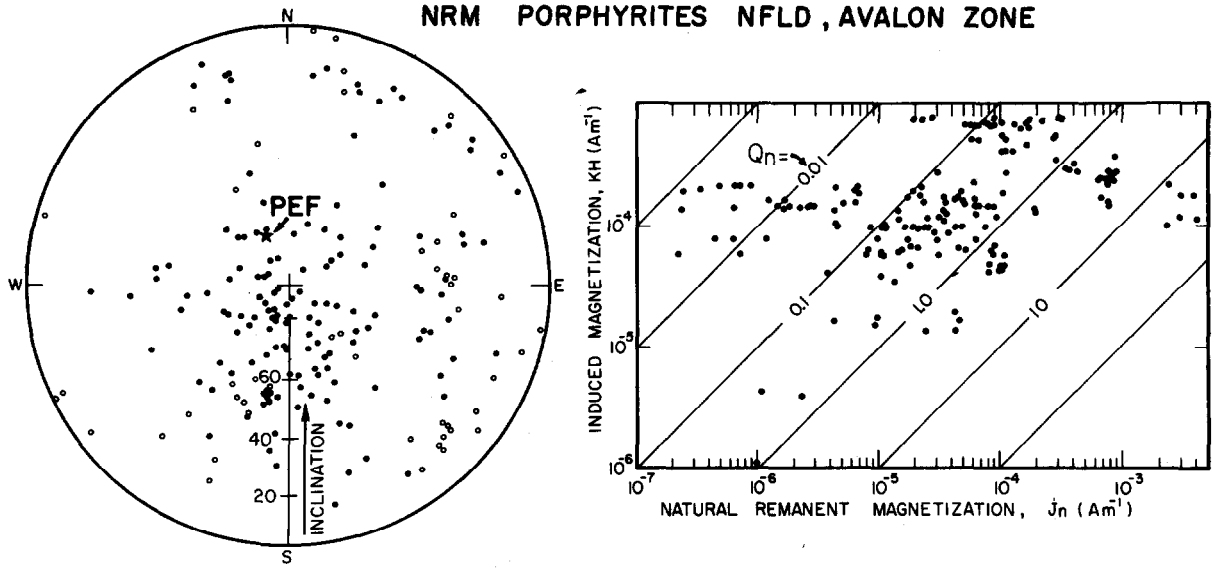


Fig. 3. Individual sample NRM directions on Wulff net for sites HB21 to HB32 (left). The solid circles represent projections on lower hemisphere. The star represents the present Earth's field (PEF) at the sampling locality. Plots of intensity of NRM (J_n in Am^{-1}) against the induced magnetization (KH in Am^{-1}) with resulting slope lines indicating the values of Koenigsberger ratio Q_n (right) for all samples.

temperature in the Earth's field ($H - 0.05$ mT), using a Cenco MS-3B susceptibility bridge. Alternating field (AF) demagnetization was carried out using a demagnetizer with a maximum peak-to-peak intensity of 90 mT in three large mu-metal cylinders (Seguin, 1975) incorporating a three-axis tumbler, and a commercial Schonstedt AF specimen demagnetizer, (model GSD-1) with a 100 mT peak-to-peak intensity. Thermal treatment of the specimens was done in a Schonstedt demagnetizer (Model TSD-1) or in an electrical oven built at University of Laval (Seguin, 1973/1974). The amount of residual ambient field present is 5 nT in the demagnetizer, and 10 nT in the electrical ovens.

Individual specimen NRM directions of porphyrite stocks and rhyolite sills (Fig. 3), although characterized by wide scatter caused by positively and negatively inclined shallow remanence and a smaller amount of steeply inclined remanence, have a pronounced deviation away from the Present Earth's Field (PEF) ($D = 335°$, $I = 71°$) in-situ. This indicates that the porphyrites and the rhyolites carry some ancient remanence especially since the directions of remanence of the few specimens (5–10%) that lie close to the PEF were found to move away from those initial directions during AF or thermal demagnetization. The majority of the NRM directions are predominantly situated in the southeast and southwest quadrants on the stereonet. The wide scatter in NRM with dual polarity of remanence is explained later in terms of the presence of both low and high coercivity components. The intensities of NRM (J_n) vary from 10^{-3} to 4 Am^{-1} with values of the Koenigsberger's ratio ($Q_n = J_n/KH$) between 0.01 and 10, with most ratios between 0.1 and 0.8 (Fig. 3). The Q_n values for both porphyrite stocks and rhyolite sills are similar; they are mainly located in the range 0.1–1.0. This observation suggests that they are probably consanguinous. The NRM orientations of the porphyrites and the rhyolites are also quite similar. The directions which are not located close to the PEF plot preferentially in the two southern quadrants and less commonly in the NE quadrant. The inclinations are predominantly downward.

6. Demagnetization studies

6.1. *Alternating field (AF) studies*

For the pilot AF study, at least six specimens per site of porphyrites and rhyolites were demagnetized in 11 steps up to 100 mT, and vector diagrams (Zijderveld, 1967) were used in the subsequent analysis. For 95% of the specimens of porphyrites and rhyolites, the intensities of magnetization at/or before the 60–80 mT treatment had fallen to less than 10% of the NRM intensity. In 30% of the specimens the measurements indicated acquisition of spurious magnetization at demagnetizing fields of 70–100 mT. Remanence behaviour during AF demagnetization typical of most specimens from porphyrite and rhyolite sites are shown by three specimens from three different sites. Vector trajectories are fairly linear until stable end-points are reached at higher fields. The linear trajectory for specimen HB22-1A displays a southeasterly directed magnetization (Fig. 4a) having shallow to intermediate upward inclination ($D = 135°$, $I = -40°$) which is present in the 15–60 mT range (more generally 40 mT). The second component has a northward directed magnetization with shallow to intermediate downward inclination ($D = 20°$, $I = +30°$) and is isolated in

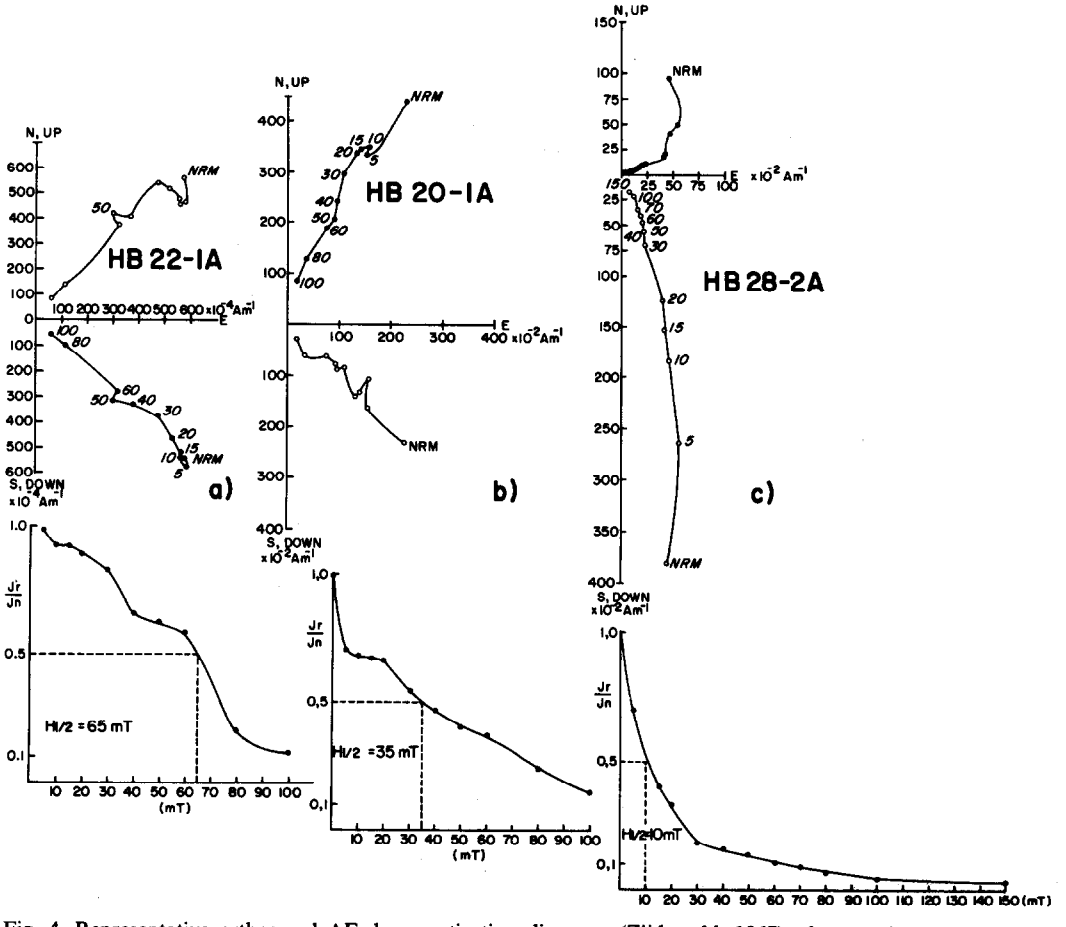


Fig. 4. Representative orthogonal AF demagnetization diagrams (Zijderveld, 1967) of successive magnetization vectors for three specimens as indicated. Solid (open) circles represent projections on horizontal (vertical) plane. The numbers next to the circles represent the demagnetization steps in mT as indicated. The respective AF decay curves are shown below. This figure illustrates an example for the isolation of the three components of magnetization.

the 30–60 mT range (Fig. 4b). This component is present on sites HB20 and 28 (porphyrites) and sites HB22 and 23 (rhyolites). A third component is isolated in the 20–80 mT coercivity range. This third component is observed on sites HB20, 28, 30 and 31 in the porphyrites and in only one site (HB26) of the rhyolites. This component has a east-northeast direction of magnetization with an intermediate to steep downward inclination (D = 70°, I = +60°). The linear trajectory of specimen HB28-2A (Fig. 4c) is a typical example of this component. A fourth component of magnetization is rarely detected and mainly in the porphyrites (sites HB21, 31 and 32). This southern direction of magnetization (with an intermediate downward inclination) is generally superimposed on the other components and is characterized by a low coercivity (10–15 mT); this situation suggests a secondary origin for this component.

6.2. *Thermal demagnetization*

At least four specimens for each of the 11 sites were thermally treated in a minimum of 12 steps up to 665°C, and vector diagrams were used in the subsequent analysis of the results. Most commonly, the unblocking temperature (T_{ub}) is located in the 500–650°C with two main modes: one at temperatures smaller than 580°C and the other around 625°C. The first T_{ub} is close to the Curie point of pure magnetite and the second one may represent an impure haematite. In 70% of the specimens the intensities of magnetization fall off to less than 10% of the original NRM intensity in the interval 550–650°C (with two modes at 570 and 640°C). Typical behaviours of three (3) porphyrite and rhyolite specimens from three different sites are shown in Fig. 5. In the case of specimen HB24-3A, the component directions were defined by a grouping of the characteristic directions taking into consideration the T_{ub}'s and the MDF's. In the case of specimen HB24-3A, a first component is detected (Fig. 5a); it is a north-northwest directed magnetization with a shallow to intermediate downward inclination (D = 310°, I = +40°). This NNW component is observed on sites HB20, 21, 28, 29, 30, 31 and 32 in the porphyrites and on sites HB22, 23 and 24 in

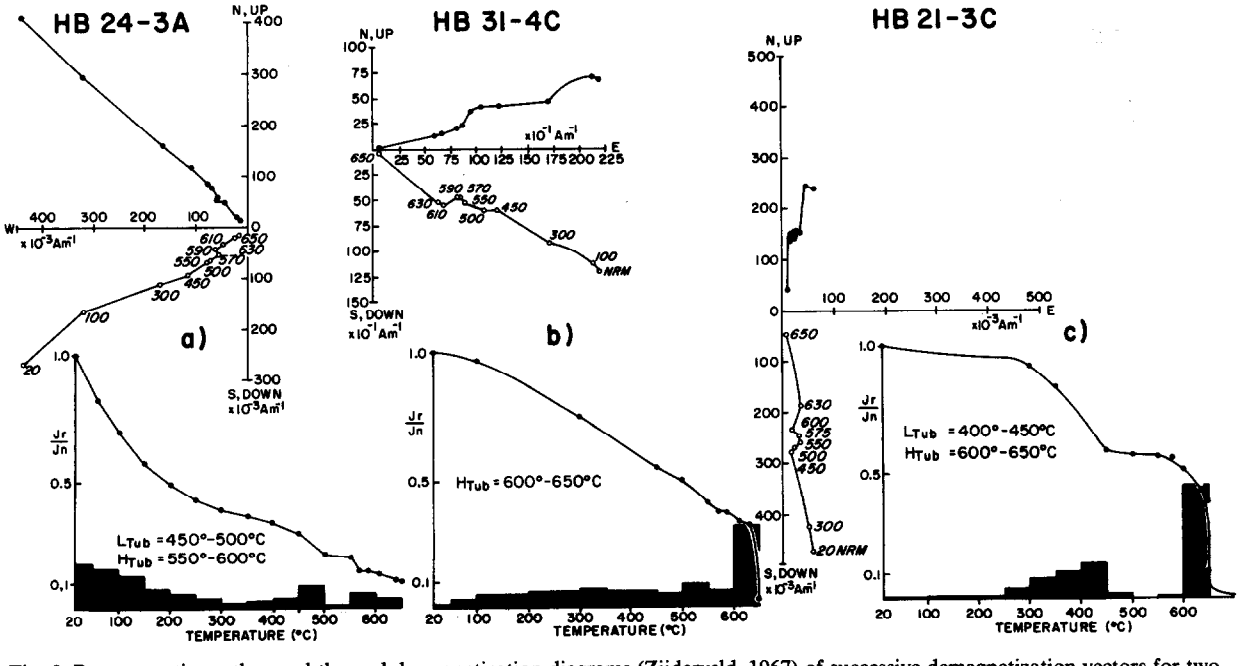


Fig. 5. Representative orthogonal thermal demagnetization diagrams (Zijderveld, 1967) of successive demagnetization vectors for two specimens as indicated. The other symbols are as in Fig. 4. The numbers next to open circles represent the demagnetization steps (°C). The unblocking temperature (T_{ub}) in °C is indicated below the orthogonal plots.

the rhyolites; it is detected at 550–650°C. A second component (Fig. 5b) is isolated at 450–610°C in sites HB20, 28, 31 and 32 in the porphyrites and at site HB26 in the rhyolites. This east-northeasterly component characterized by a shallow to intermediate downward inclination ($D = 70°$, $I = +35 \rightarrow 40°$) is typically represented by specimen HB31-4C. A northeast directed component of magnetization with an intermediate to steep downward inclination ($D = 15°$, $I = +50°$) is represented by an atypical specimen (HB21-3C; Fig. 5c). This component is best isolated in the temper-

TABLE I

Statistical summary of site–mean directions

Site no.	No. of samples	No. of specimens	Component no.	P	D (°)	I (°)	α_{95} (°)	k (°)	AF range (mT)	T range (°C)
			Porphyrites							
HB-20	3	4	A	N	11	18	15.5	61	20–100	
	2	5	B	N	67	55	–	–	30–40	450–610
	2	4	C	N	348	23	–	–	20–50	550–600
HB-21	1	2	A	R	8	56	–	–		575–600
	3	6	C	N	273	50	31	16	20–60	550–630
	3	5	D	M	164	53	30	18	20–50	575–630
HB-28	1	4	A	R	7	50	–	–	40–100	550–630
	3	6	B	M	68	55	39	9	10–100	600–630
	4	9	C	M	304	59	27	11	20–100	500–650
HB-29	5	14	C	R	329	55	15	25	20–100	500–630
HB-30	2	5	B	N	41	42	–	–	30–100	
	4	12	C	R	326	59	13.5	41	60–80	550–650
HB-31	1	3	B	M	74	32	–	–	25–90	570–630
	4	7	C	N	294	49	22	19	70–100	575–650
	4	10	D	N	194	45	26	14	25–60	450–580
HB-32	2	3	B	N	86	38	–	–		550–575
	4	6	C	N	306	51	22	18	50–80	630–650
	5	13	D	N	191	36	17	22	20–50	550–630
	No. of sites									
	3/7		A	M	9	41	31	17		
	5/7		B	M	67	45	15	27		
	7/7		C	M	313	52	15	16		
	3/7		D	M	184	45	21	35		
			Rhyolites							
HB-22	2	3	A	N	11	35	–	–	60–100	550–575
	4	10	C	M	305	45	9.4	95	50–80	570–610
HB-23	2	4	A	N	12	46	–	–	20–60	570–610
	4	9	C	M	292	47	35	8	40–80	590–630
HB-24	2	2	A	N	25	16	–	–		630
	5	10	C	M	322	44	32	8	25–80	550–630
HB-26	2	3	B	M	52	41	–	–	20–50	550–600
	4	10	D?	N	147	51	44	6	30–60	550–630
	No. of sites									
	3/4		A	N	17	33	24	27		
	3/4		C	M	307	46	16	59		
	Porphyrites + rhyolites									
	6		A	M	13	37	14	22		
	10		C	M	311	50	11	21		

ature range 550–630°C and is observed at sites HB21 and 28 in the porphyrites and sites HB22, 23 and 24 in the rhyolites. In the case of specimen HB21-3C and a few others having a similar behaviour, the characteristic component direction is obtained in the 450–630°C range because at 650°C the magnetic susceptibility increased substantially indicating the appearance of an artificial magnetic mineral. In the 450–630°C range, the NRM intensity does not decrease much and it is difficult to estimate the change in declination; a straight line regression using the least mean square method yields a northeast direction. A less common fourth component detected at sites HB21, 31 and 32 in the porphyrite only is best isolated in the temperature range 450–630°C. Its direction of magnetization is S with an intermediate downward inclination.

In at least 80% of the cases, the AF demagnetized and thermally treated components coincide fairly well. This is not so when both haematite and magnetite are present in the same specimen because of the different magnetic hardness of the two carrier minerals. Occasionally, both normal and reverse directions are observed in the course of thermal demagnetization. Some specimens have shown an increase of NRM intensity in the temperature range: 590–630°C (average: 615°C) suggesting either mineralogical change(s) in the original magnetic memory carrier or opposite polarities in the magnetite and haematite carriers. AF demagnetization is probably less efficient than thermal treatment for most of the sites.

7. Statistical analysis of site-mean directions

Site-mean directions for combined AF and thermally demagnetized results are presented in Table I. North-northeast(A), east-northeast(B), northwest(C) and south(D) remanence directions are observed in either porphyrite stocks or rhyolite dykes and the corresponding group-mean directions based on sites are given in this Table. A, B and C directions are common to both porphyrites and rhyolites. Component D is observed only in the porphyrites. The PEF component is almost absent in the demagnetized results except in two specimens of site HB 32. As a result of these observations, only the A, B and C components are common to both porphyrite intrusive and rhyolite dyke sites and are thus considered reliable for palaeomagnetic investigation.

8. Sequence of the components of magnetization

An attempt to classify the acquisition sequence of these components was based on: (1) the coercivity spectrum and unblocking temperature, (2) the frequency distribution (number of specimens and samples per component), and (3) the precision parameter (k) of the mean direction. Based on criterion (1) applied to specimens or samples of the different porphyrite and rhyolite sites, it is assumed that the components having the higher coercivity spectra (H_{cr}) and higher unblocking temperatures (T_{ub}) are the oldest. Clearly, this is not always true since the H_{cr} and the T_{ub} depend on the nature of the magnetic carriers. By working out this relationship, the following sequence was established: C > A > B. The sequence C > A > B holds for both AF and thermally demagnetized specimens. On the basis of frequency distribution (number of specimens per component) for both porphyrite stocks and rhyolite dykes (see Table I), the sequence is the following: C > A $\geqslant$ B. Finally, when using the largest precision parameter (k) as a criterion, the sequence arrived at is: C > A $\geqslant$ B for the porphyrites and rhyolites combined. These three criteria point out to the same acquisition sequence of magnetization: Component C is older than component A which is in turn older than component B. As these approaches are indirect, the conclusions can be only tentative.

9. Palaeopole positions and their relation to poles previously determined

The C component ($D = 313°$, $I = +52°$) yields a palaeopole at 48°N 212°E, the A component ($D = 9°$, $I = +41°$) a palaeopole at 61°N 101°E and the B component ($D = 67°$, $I = +45°$) a palaeopole at 34°N 33°E (see Table II). The D component ($D = 184°$, $I = +45°$) has given a palaeopole of 16°S 57°W. In the section dealing with the sequence of acquisition of magnetization,

TABLE II

Late Hadrynian palaeopoles of the North American craton and Avalon

Rock unit and location	Pole position Age (Ma)	Lat. (°)	Long. (°)	d_p (°)	d_m (°)	α_{95} (°)	Pole	Reference
	Avalon Zone							
Blackhead Fm, Nfld	~ 600	5 S	96 W	9.1	13.4		BH	Nairn et al. (1959)
Signal Hill Gp., Nfld	~ 650	16 N	142 W	6.2	11.8		SH	Nairn et al. (1959)
Finn Hill Sequence, Nfld	(A) ~ 590	14 S	1 W			8	FH(A)	Seguin (1985)
Finn Hill Sequence, Nfld	(B) ~ 750	39 S	29 E			30	FH(B)	Seguin (1985)
Porphyrite and rhyolite, Nfld	A	61 N	101 E	9.6	16.4			This study
Porphyrite and rhyolite, Nfld	B	34 N	33 E	12	19			This study
Porphyrite and rhyolite, Nfld	C	48 S	22 E	9.8	14.7			This study
	Cratonic North America							
Victoria Island	~ 630	2 N	161 E	–	–		VI	Palmer et al. (1983)
Franklin Intrusive 1	~ 665	5 N	165 E			5	FI	Irving and Hastie (1975)
Franklin Intrusive 2	~ 665	6 N	166 E			4	FI	Irving and Hastie (1975)
Franklin Intrusive 3	~ 665	5 N	165 E			3	FI	Irving and Hastie (1975)
Cloud Mt. Basalt, Nfld	~ 605	5 N	172 E	2	4		CM	Deutsch and Rao (1977)
Rapitan Gp. X	~ 760	7 S	194 E			5	RX	Morris (1977)
Rapitan Gp. Z	~ 640	5 N	149 E			4	RZ	Morris (1977)

it was concluded that component A is secondary, i.e., Palaeozoic or even Mesozoic. The palaeopole of component A is not at all close to other early Palaeozoic poles from cratonic North America or northwestern Europe after closure of the present Atlantic Ocean but is near Late Carboniferous and Early Permian ones, indicating that the porphyrite stocks and rhyolite sills were remagnetized in the late Palaeozoic. Component B appears to be younger than component A according to the conclusions arrived at in the previous section. However, the latitude of the B pole is lower than that of the A pole. This could mean that the magnetization of component B is imperfectly isolated and still unresolved, somewhere midway between the A and C components of magnetization.

Component B has a larger circle distribution than components A and C, but it should be possible to resolve it as A and C are of different magnetic hardness. Component B could be the result of a specific tectonic event; however, there is no observed evidence of Palaeozoic dislocation of the porphyrite stocks and rhyolite sills. A comparison of the mean direction of component B with mean directions obtained from early Cambrian (Brigus shale) and Hadrynian basaltic and mafic flows was attempted. The west, intermediate positive direction of Brigus shale is definitely unrelated to the direction of component B. One unfrequent component (D) encountered at 2/10 sites in the Hadrynian flows yielded a direction of approximate reverse polarity (266, −37) of component B. Component D is characterized by a negative fold test and is definitely secondary. Thus, component B is most probably an overprint of chemical nature. The antipole of component C (31°E, 48°S) is

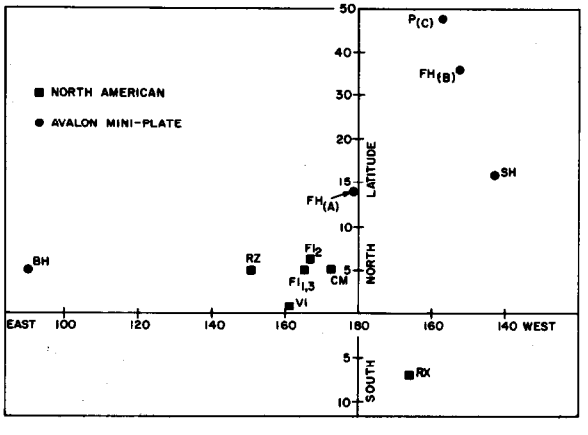


Fig. 6. Palaeomagnetic poles (late Hadrynian) derived from Avalon Zone and cratonic North America (Table II).

comparable to pole B (29°E, 39°S) of the Finn Hill Sequence (Seguin, 1985; Table II) which is interpreted as primary. As the porphyrite intrusives and rhyolite sills are contemporaneous or slightly younger than the Finn Hill ignimbrites, the C magnetization of the porphyrites and rhyolites is most likely to be primary.

The palaeopoles obtained for the Cloud Mountain basalt (Deutsch and Rao, 1977) and the Black Head Formation (BH) and Signal Hill (SH) Group (Nairn et al., 1959) are 5°S, 8°W; 5°S, 96°W and 16°S, 38°E, respectively (Fig. 6). Only the two last rock units and the Finn Hill Sequence belong to the Avalon Zone; no specific trend for an APW trajectory is apparent yet. The inclusion of the SH and BH poles in the APW path is unjustified because of the poor reliability of these poles (E.R. Deutsch, personal communication, 1985). Thus, only the Finn Hill sequence pole and that presented in this study can be used to construct a tentative Hadrynian APW path for the Avalon microcontinent and to establish a suitable relation with the North American craton and western Eurasia. Pole $P_{(C)}$ in this study coincides quite well with pole $FH_{(B)}$ characterized by a primary component (Fig. 6). The significant difference between Avalon poles and possible contemporary poles from North America (Table II) imply the presence of separate plates at this time.

These new palaeomagnetic constraints added to existing geological constraints allow comparison between the Avalonian and other Atlantic-bordering terranes, especially Armorica and the northwest Europe–Baltic plate. A possible scenario of plate tectonics motion of Avalon in Hadrynian time is discussed in Seguin (1985). The only significant difference in this investigation is the detection of the superimposed Alleghenian orogenic event.

Acknowledgements

The authors thank Dr. E.R. Deutsch and T. Vallis (Earth Sciences Department, Memorial University) for assistance in drilling and cutting of oriented rock samples, K.V. Rao (Université Laval) for field assistance and R. Ouellet and James Jessop (Department of Physics, Université Laval) who made some of the measurements. This work was supported by EMR research agreement (Canada) No.: 128-3-80 and NSERC (Canada) No: 7070-1980-81-82. Two unknown referees contributed largely to the improvement of the manuscript.

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